

Reasoning + Acting

Fundamental limitation of LLM :

- LLMs can generate answers in the form of tokens.

- But they cannot verify everything.

If we ask a LLM to fix a bug in a code repo, then it might already know the prog lang, the db used, the other part of the tech stack, but it doesn know what your code repository is actually contains.

Tools ->

HITL

Very simple coding agent -> Claude Code (MVP1)

Long running tasks -> Message queues

What set of tools might be useful for us to run this claude code ?

-> A general developer does creation/editing/deletion of code files and try to run them .

-> Tool -> Create New files., Edit existing files, Delete files, List files by name regex etc. Filesystem

-> Tool -> A tool which can give our agent access to run bash / cli commands.

python3 main.py

-> Tool -> read file

Bug -> Log Stack ---> agent

Act

CoT -> Some level of reasoning -> Reasoning based models.

-> 01 , r1 deepseek

-> intermediate set of token

Write a backend server in python for me with a hello world route

-> from flask import xyz .....

Question -> Reason -> Action -> Observation -> Reason ..... -> Answer

Write a ci cd pipeline to verify all the migrations in the db are valid ?

-> What ci cd tool we are using in the repo ?

-> What language/framework the repo is in ?

-> What migration library it is currently ?

-> Are there existing ci cd workflows to check migrations

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-> Modern agents tend to verify critical decisions from you. How is this implemented?

-> HITL -> Human in the loop

HW: Try to prepare a simple coding agent using plain python agent loop.

-> It should be able to to do crud on filesystem and run commands on your terminal.

-> Two set of commands:

1. Blacklisted -> No go -> your agent should not run it

2. Having some side effect: -> Your agent shoudl evaluate if the command is possible to create a side effect and if yes then trigger hitl